

ELASTICITY PROBLEM SET—SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

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PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

1 Warm up Exercise: Calculating Elasticities

- (a) $D(p) = a - bp. \Rightarrow \frac{dD(p)}{dp} = -b \Rightarrow \varepsilon_D^p = -\frac{bp}{a-bp}.$
- (b) $D(p) = 60 - p. \Rightarrow$ same as above, with $b = 1 \Rightarrow \varepsilon_D^p = \frac{-p}{a-bp}.$
- (c) $D(p) = \frac{40}{p^2}. \Rightarrow \frac{dD(p)}{dp} = -2 \left(\frac{40}{p^3} \right) \Rightarrow \varepsilon_D^p = -2 \left(\frac{40}{p^3} \right) \left(\frac{p}{40/p^2} \right) = -2.$
- (d) $D(p) = Ap^{-b}. \Rightarrow \frac{dD(p)}{dp} = -bAp^{-b-1} \Rightarrow \varepsilon_D^p = -bAp^{-b-1} \left(\frac{p}{Ap^{-b}} \right) = -bp^{-b-1}p^{b+1} = -b.$
- (e) $D(p) = (p+3)^{-2}. \Rightarrow \frac{dD(p)}{dp} = -2(p+3)^{-3} \Rightarrow \varepsilon_D^p = -2(p+3)^{-3} \left(\frac{p}{(p+3)^{-2}} \right) = -\frac{2p}{3+p}.$

2 Another warm-up exercise

- (a) From problem 1, the elasticity of demand is $\varepsilon_D^p = \frac{-p}{240-p}.$
- When the $p = 0$, $\varepsilon_D^p = 0.$
 - If $p \rightarrow 240$ then the elasticity tends to negative infinity, i.e., $\lim_{p \rightarrow 240} \frac{-p}{240-p} = -\infty.$ Thus, when p approaches 240, the demand becomes infinitely elastic.
 - Solve $\frac{-p}{240-p} = -1$ to get that at the price of $p = 240/2 = 120$, the elasticity is unitary.
 - $\frac{-40}{240-40} = -0.2.$
- (b) The demand for Pokemon cards is given by

$$Q_d = 500,000 - 45,000p$$

- $Q_d(2.5) = 500,000 - 45,000(2.5) = 387,500.$
- $-\frac{45000(5)}{500000-45000(5)} = -\frac{9}{11}.$

3 Elasticity, Total Revenue and Arc-Elasticity

First, let us rewrite the demand and supply functions as $Q_D = 2200 - 100P$ and $Q_S = 200P - 800$, respectively. So the equilibrium is reached at the point where $2200 - 100P = 200P - 800 \Rightarrow 300P = 3000 \Rightarrow P^* = 10$ and thus $Q^* = 1200.$

- (a) The elasticity for demand is equal to

$$\varepsilon_D = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = -100 \cdot \frac{P}{2200 - 100P} = \frac{-100P}{2200 - 100P}$$

At equilibrium, i.e. at $P = 10$ the elasticity is

$$\varepsilon_D = \frac{-1000}{1200} \approx -0.833$$

(b) The elasticity for supply is

$$\varepsilon_S = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = 200 \cdot \frac{P}{200P - 800} = \frac{200P}{200P - 800}$$

At equilibrium, i.e. at $P = 10$, the elasticity is

$$\varepsilon_S = \frac{2000}{1200} \approx 1.667$$

(c) Since the demand is inelastic at equilibrium ($|\varepsilon_D| < 1$), after any small change in price, the percentage change in quantity demanded will be less than the percentage change in price, so the expression PQ , which is the revenue, will change in the same direction as price (i.e., if price increases, revenue increases and vice-versa)

(d) Let $P_1 = 10$ and $P_2 = 12$, so corresponding quantities demanded are $Q_1 = 1200$ and $Q_2 = 1000$ respectively. So $\bar{P} = \frac{10+12}{2} = 11$ and $\bar{Q} = \frac{1200+1000}{2} = 1100$. The arc-elasticity between these two points is:

$$\varepsilon_{1,2} = \frac{\Delta Q}{\Delta P} \cdot \frac{\bar{P}}{\bar{Q}} = \frac{-200}{2} \cdot \frac{11}{1100} = -1$$

4 The market for cat treats

(a) For cat cafes, the elasticity is

$$\varepsilon_1 = \frac{\Delta q_1}{\Delta p} \cdot \frac{p}{q_1} = -1 \cdot \frac{p}{200 - p} = \frac{-p}{200 - p}$$

and for cat owners, the elasticity is

$$\varepsilon_2 = \frac{\Delta q_2}{\Delta p} \cdot \frac{p}{q_2} = -4 \cdot \frac{p}{90 - 4p} = \frac{-4p}{90 - 4p}$$

(b) The elasticity equal to -1 means

$$\frac{-p}{200 - p} = -1 \implies p - 200 = -p \implies p = 100$$

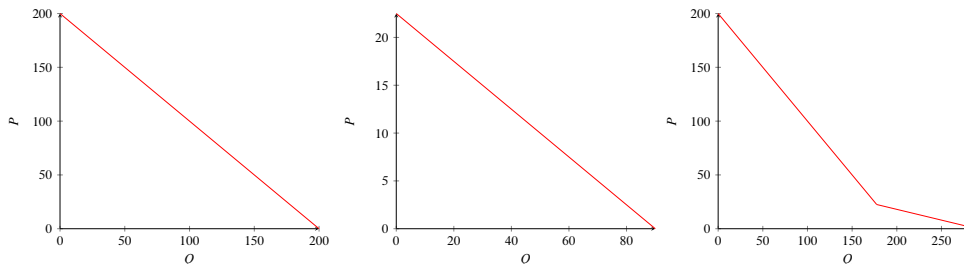
(c) The elasticity equal to -1 means

$$\frac{-4p}{90 - 4p} = -1 \implies -4p = 4p - 90 \implies 8p = 90 \implies p = \frac{45}{4}$$

(d) First let's note that market demand is described by the following equation:

$$Q = \begin{cases} 290 - 5p & \text{if } p \leq \frac{90}{4} \\ 200 - p & \text{if } p > \frac{90}{4} \end{cases}$$

Thus, the diagrams for the cat cafes' demand, the cat owners' demand, and market demand are the following:



(e) One can see that the elasticity for market demand is described by the following equation:

$$\varepsilon = \begin{cases} \frac{-5p}{290-5p} & \text{if } p \leq \frac{90}{4} \\ \frac{-p}{200-p} & \text{if } p > \frac{90}{4} \end{cases}$$

So the elasticity is equal to -1 either if $\frac{-5p}{290-5p} = -1$ and $p \leq \frac{90}{4}$ or if $\frac{-p}{200-p} = -1$ and $p > \frac{90}{4}$. It is easy to see that the first condition is not satisfied for $p \leq \frac{90}{4}$, so the only such point is $p = 100$, and revenue is maximized at this point.

5 Income and cross-price elasticities

- (a) Quantity demanded also decreases by $2\% \cdot 0.7 = 1.4\%$.
- (b) The demand for skies decreases by $8\% \cdot 1.5 = 12\%$.
- (c) The demand for increases by $20\% \cdot 1.5 = 30\%$.